Tony Tang

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Education

Massachusetts Institute of Technology, Cambridge, MA September 2023 – May 2027
Bachelor of Science in Mechanical Engineering GPA: 4.7/5.0
Relevant Coursework: Design and Manufacturing I-II, Thermodynamics, Electronics for Mechanical Systems, Mechanics and Materials I, Dynamics and Control I-II, Mobile Autonomous System Lab, Numerical Computation

Experience

Intern | SUL, San Diego, CA September 2025 –July 2026

- Prototyped and assembled 3 machine modules for an automated food-packaging machine using 3D-printed parts, sheet metal, conveyor hardware, and other commercially available materials.
- Developed the electrical system and control logic to coordinate dispensing, conveyor motion, and continuous bowl flow based on user input.
- Engineered an improved food-dispensing mechanism that increased packaging throughput by 50% while maintaining controlled portioning.

Intern | SINTEF Ocean, Trondheim Norway June 2025 – August 2025

- Created an URDF model of the Reach Alpha5 manipulator using SolidWorks for ROS2 integration
- Incorporated MoveIt into RViz to construct a simulation where the Reach Alpha5 Manipulator executed motion planning from Cartesian coordinates via inverse kinematics
- Debugged ROS2 systems using rqt_grapt

Undergraduate Research | MIT Sea Grant, Cambridge, MA February 2024 – May 2024

- Designed and developed a handheld controller for the Wave Generator Project that adjusts the distance and speed of a linear actuator, regulating the velocity and frequency of the waves.
- Utilized Arduino Uno and electrical components such as motor driver, potentiometer, and switch to develop a user-friendly controller to control the waves

Activities & Extracurriculars

Mechanical Engineer | Autonomous Robotics Team, Cambridge, MA September 2023 – January 2026

- Developed thruster cages for an autonomous vehicle using CAD software and machining tools, enabling durability to withstand over 200 pounds.
- Constructed hulls for a catamaran using Styrofoam and epoxy materials, optimizing weight for competitive performance
- Design a waterproof and modular electronics box with organized component layout and enhanced accessibility features.

Discover Ocean Engineering Counselor | MIT Sea Grant, Cambridge MA August 2024, August 2025

- Guided 30 freshmen through a comprehensive orientation program, showcasing MIT's ocean engineering facilities, campus highlights, and Boston landmarks. Provided insights into the ocean engineering curriculum and student experience at MIT.
- Managed a $14,000 budget for program activities and resources.
- Supervised students in the design and construction of competitive underwater battle robots, fostering hands-on engineering skills and teamwork.

Skills

Hardware: Design and Manufacturing, Machine Operation (mills, lathes, CNC mill, 3d printers, injection molding, thermoforming, etc.), and Circuit Testing (oscilloscopes and multimeters)
Software: MATLAB, Fusion 360, SolidWorks, LTspice, Python, and Arduino IDE